

Civics Group	Index Number	Name (use BLOCK LETTERS)	H2
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ST. ANDREW'S JUNIOR COLLEGE
2022 JC2 PRELIMINARY EXAMINATIONS

H2 BIOLOGY

Paper 2

Monday 29th August 2022 Total 2 hours

Materials: Question Papers

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

The number of marks is given in brackets [] at the end of each question or part question.

Conceptual error (C)	Data Quoting (D)	Expression (E)	Misreading the question (Q)

For Examiners' Use	
1	/10
2	/10
3	/8
4	/12
5	/10
6	/10
7	/10
8	/11
9	/8
10	/6
11	/5
Total	/100

This document consists of 32 printed pages.

[Turn over]

QUESTION 1

(a) Fig. 1.1 shows an electron micrograph of a plant cell.

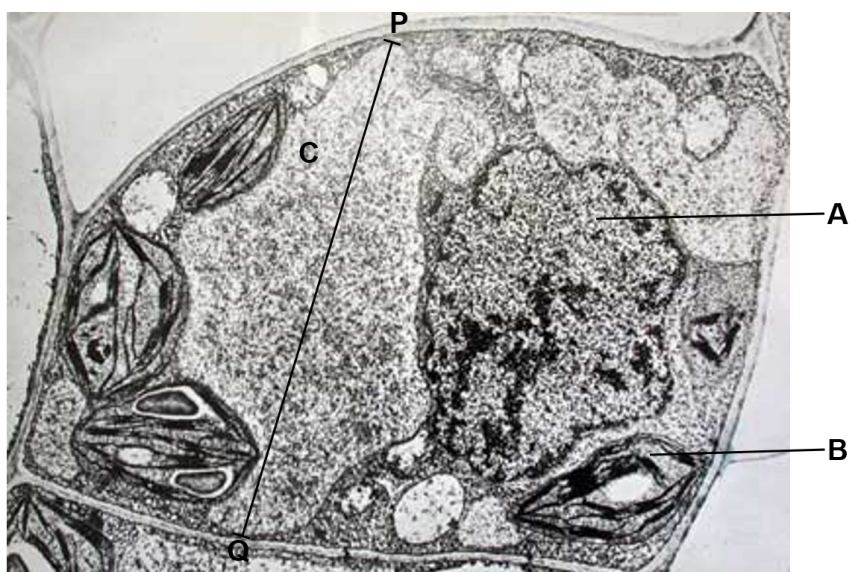


Fig. 1.1

- (i) The magnification of the photomicrograph is x560. Calculate the actual length of organelle **C** in μm , along the line **P-Q**. Show your working.

Actual length of organelle **C**: μm [2]

- (ii) Identify organelles **A** and **B** and compare their structural features.

Organelle **A**: Organelle **B**:

.....[4

- (b) Cyclins are a group of proteins that regulate the cell cycle. There are four basic types of cyclins found in humans and most other eukaryotes: G₁ cyclins, G₁/S cyclins, S cyclins, and M cyclins.

Each cyclin is associated with a particular phase, or set of phases, in the cell cycle and helps drive the events of that phase or period. For instance, M cyclin promotes the events of M phase.

The levels of the different cyclins vary considerably across the cell cycle, as shown in Fig. 1.2.

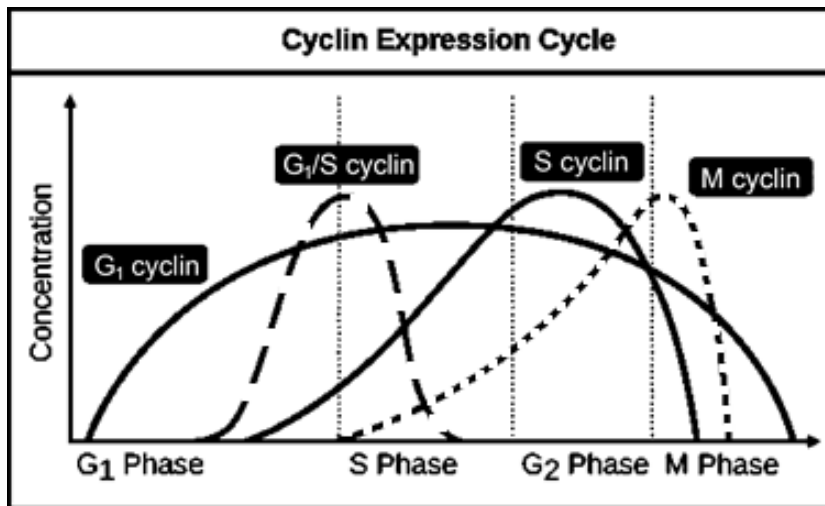


Fig. 1.2

A typical cyclin is present at low levels for most of the cycle but increases rapidly at the stage where it is needed. In order to drive the cell cycle forward, a cyclin must activate enzymes called the cyclin-dependent kinases (Cdks). In general, Cdk levels remain relatively constant across the cell cycle, but Cdk activity changes as levels of the various cyclins rise and fall.

Fig. 1.3 shows the cell cycle checkpoints and the activation periods of various Cdks. Percentage times (%T) in each phase are approximations of total cell cycle time. For instance, cells spend approximately 5% of their total cell cycle time in M phase.

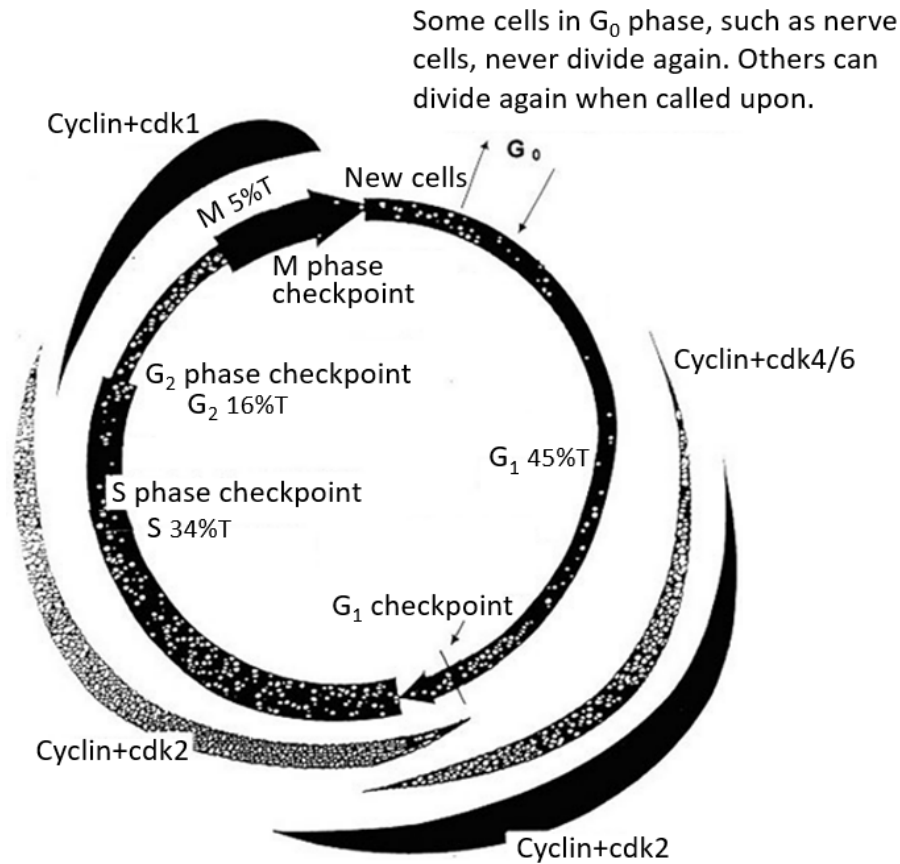


Fig. 1.3

- (i) With reference to post-translational control of gene expression, suggest how cells reduce the concentration of cyclins when they are not needed in the cell cycle.

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 [1]

- (ii) With reference to Fig. 1.2 and 1.3, identify the molecules that are at abnormally low concentrations for cells to be arrested at S phase.

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- (iii) It was found that cancerous cells show only 2%T at G2 phase. Explain the implications of this.

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[Total: 10]

QUESTION 2

- (a) Explain how the molecular structure of haemoglobin is related to its functions.

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(b) Blood arriving in the lungs from the heart is oxygenated as it passes through the pulmonary capillaries. Sickle-shaped red blood cells are present in a person with sickle cell anaemia. These red blood cells have a very high quantity of abnormal haemoglobin and take up and transport less oxygen than red blood cells containing normal haemoglobin.

The cause of the differences between abnormal haemoglobin and normal haemoglobin is a mutation in the gene that codes for one of the two types of polypeptide found in a haemoglobin molecule. This mutation leads to a change in the mRNA produced during transcription, causing a change in the primary structure of the polypeptide formed.

Fig. 2.1 shows some of the changes that occur as a result of this gene mutation.

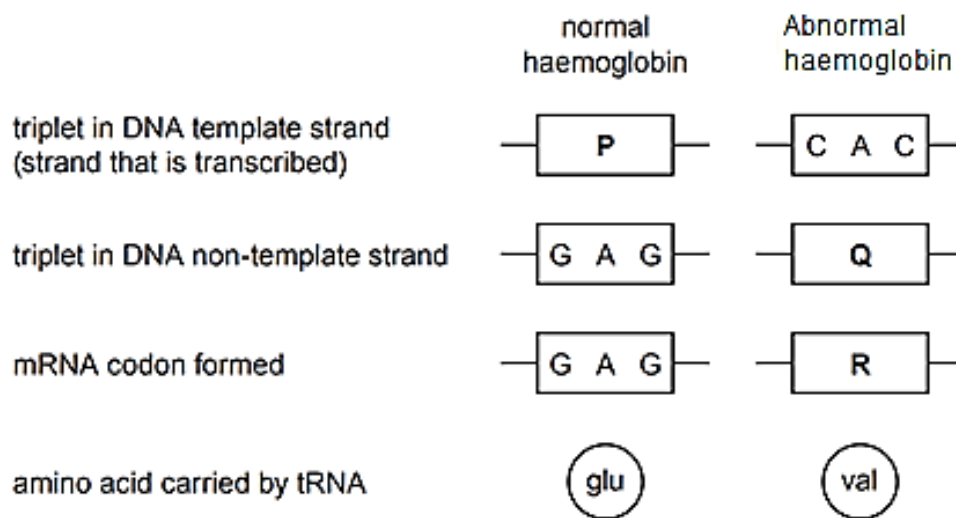


Fig. 2.1

(i) With reference to Fig. 2.1, state:

- the base sequence of DNA triplet P

.....[1]

- the base sequence of DNA triplet Q

.....[1]

- the base sequence of mRNA codon R

.....[1]

(ii) Name the polypeptide that is altered in the abnormal haemoglobin molecule.

.....[1]

(c) Fig. 2.2 shows the oxygen dissociation curve for adult haemoglobin in a person who does not have sickle cell anaemia.

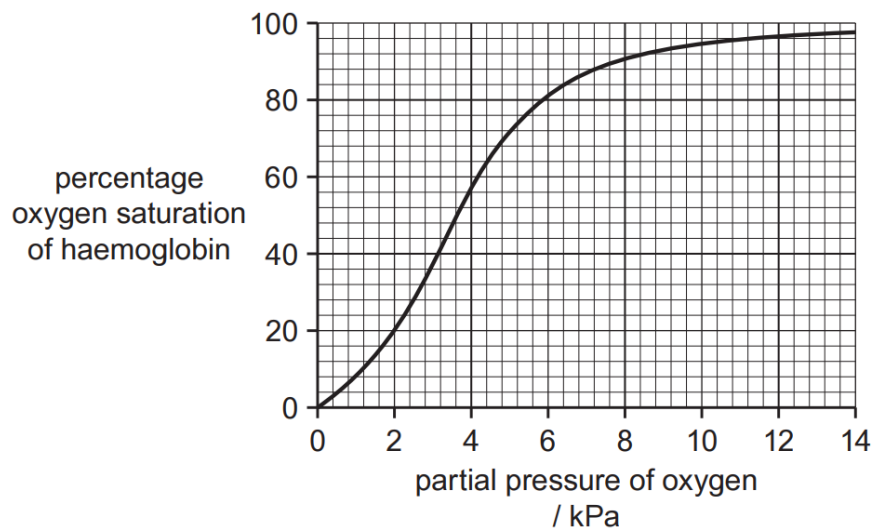


Fig. 2.2

Compared to Fig. 2.2, the oxygen dissociation curve for adult haemoglobin in a person with sickle cell anaemia is shifted to the right. The uptake of oxygen by haemoglobin in the lungs and the release of oxygen by haemoglobin in respiring tissues is different in a person with sickle cell anaemia compared with a person who does not have the disease.

With reference to Fig. 2.2, state **and** explain these differences.

Uptake of oxygen

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release of oxygen

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Explain these differences.....

.....[3]

[Total: 10]

QUESTION 3

HER2-positive breast cancer is a breast cancer that tests positive for a protein called human epidermal growth factor receptor 2. A member of the HER family of tyrosine kinase receptors (HER1–4), HER2 is an essential breast cancer oncogene.

Fig. 3.1 shows the Southern blot (DNA analysis) and Western blot (protein analysis) results of HER2 gene expression for a healthy individual and a breast cancer patient, and Fig. 3.2 shows the HER2 signaling pathway.

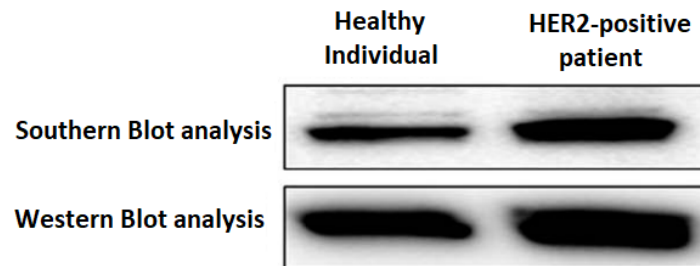


Fig. 3.1

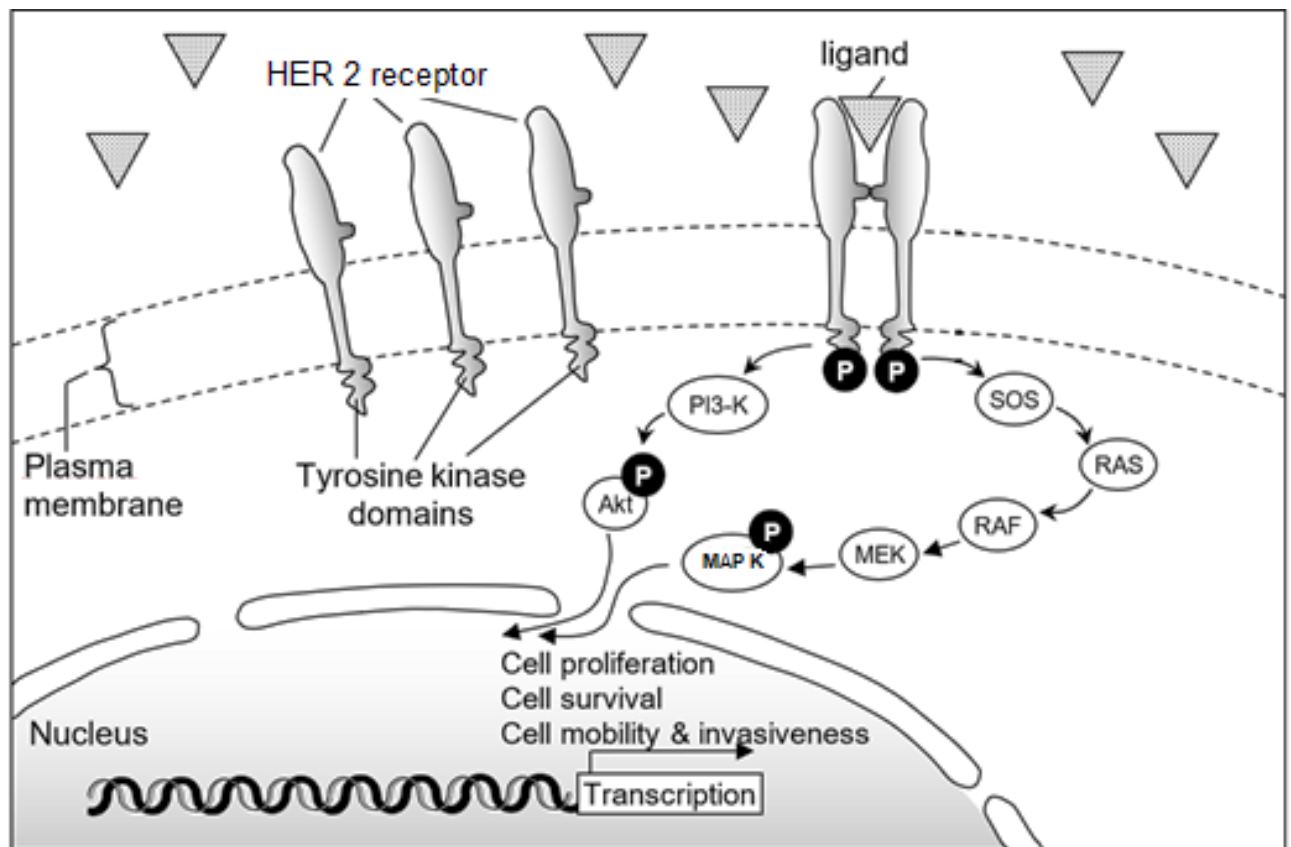


Fig. 3.2

- (a) Research shows that 1 in 5 cancer patients shows an over-expression of HER2 gene. With reference to Fig. 3.1, suggest how the HER2 gene is converted to an oncogene in cancer patients.

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- (b)** Distinguish between how the HER2 receptor and a G-protein linked receptor function as transmembrane receptors upon binding of their respective ligands.

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- (c)** Patients with HER2-positive breast cancer have responded well to treatment when the drug Herceptin is used in combination with other chemotherapy drugs. Herceptin binds to the extracellular domain of the receptor.

With reference to Fig. 3.2, suggest and explain how Herceptin may act to prevent further progression of breast cancer.

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- (d) Even with the effective treatment of Herceptin and chemotherapy, a proportion of cancer patients who were in remission, still experienced recurrence of cancer some years later.

Suggest how the cancer cells in these cases overcome the effects of Herceptin.

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[Total: 8]

QUESTION 4

- (a) Fig. 4.1 shows the small ribosomal subunit rRNA loops containing double-stranded sections, H27, H30, H31 and H34, from bacteria and the mitochondria of mammals and plants. Note: Grey regions highlights the locations of H31 and H34.

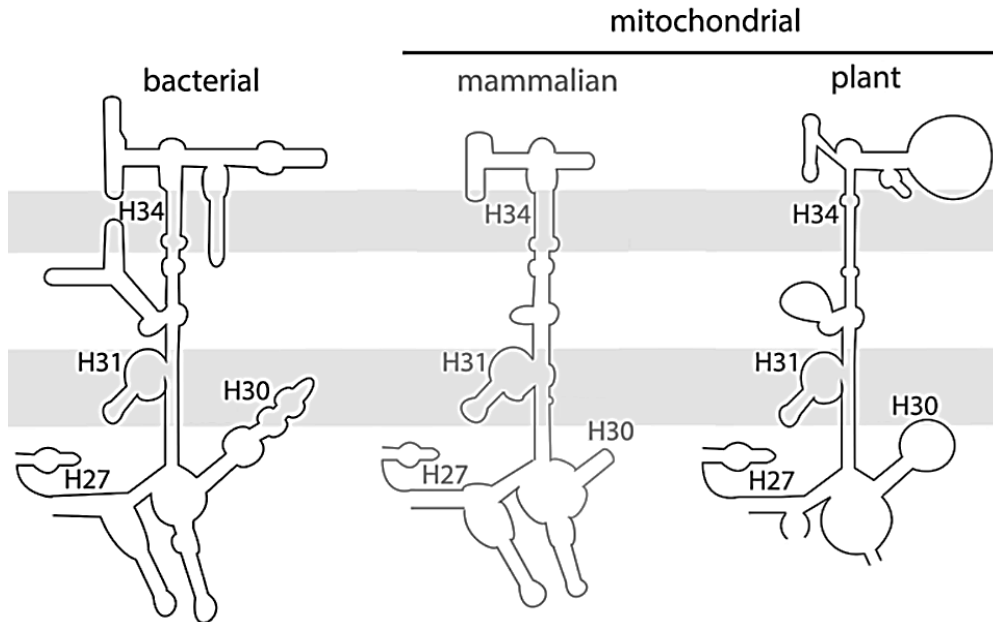


Fig. 4.1

- (i) With reference to Fig. 4.1, compare the structures of the bacterial and mammalian mitochondrial small ribosomal subunit rRNA.

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- (ii) Describe the roles of rRNA in the process of protein synthesis.

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- (iii) Mammals and plants belong to the domain Eukarya while bacteria are grouped into their own domain, Bacteria. With reference to Fig. 4.1, comment on the suitability of using the small ribosomal rRNA structure to study the phylogenetic relationships between bacteria, mammals and plants.

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- (b) Fig. 4.2A shows an electron micrograph of DNA replication occurring in organism X. Fig. 4.2B shows a representative diagram of the same process.

The dotted line represents the daughter strands. Ori R represents the origin of replication.

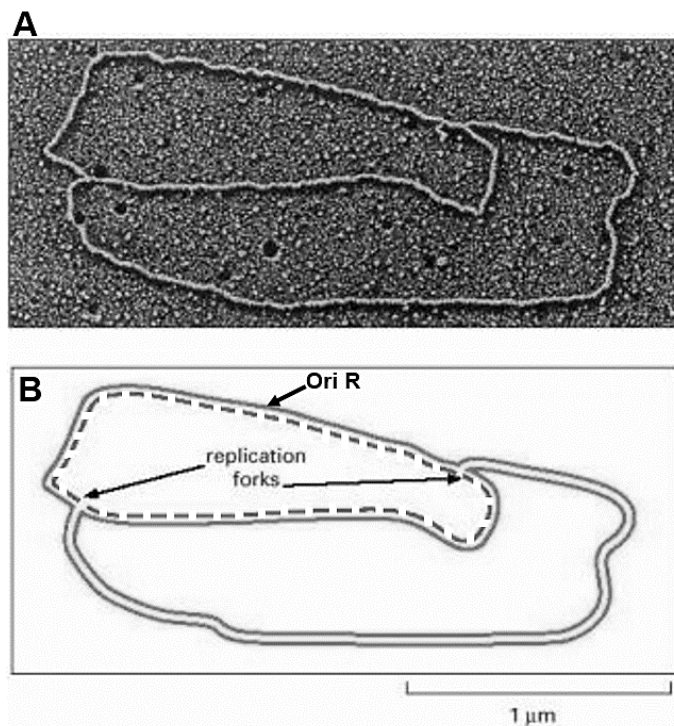


Fig. 4.2

- (i) State whether organism X is a prokaryote or a eukaryote. Explain your answer.

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 [2]

Fig. 4.3 shows a diagrammatic representation of the DNA replication process. The arrow in the diagram on the left shows the direction of the synthesis of one of the daughter strands.

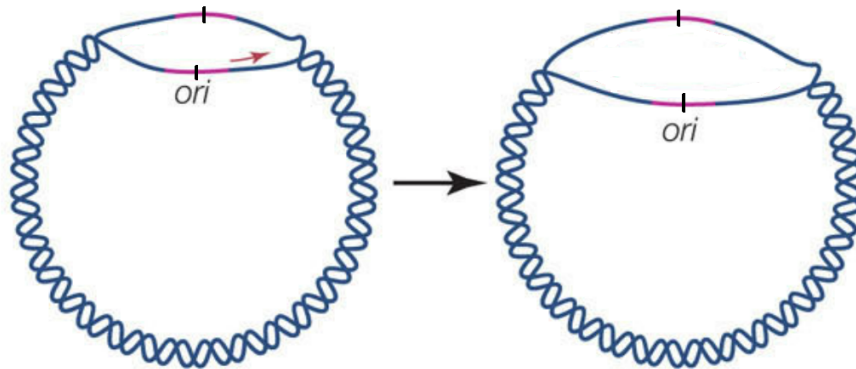


Fig. 4.3

- (ii) Draw arrows in Fig. 4.3, on the DNA molecule to the right, to indicate how the Okazaki fragments are synthesised. [1]
- (iii) Describe how the process of DNA replication differs between prokaryotes and eukaryotes.

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 [2]

[Total: 12]

QUESTION 5

Gibberellins (GA) are plant hormones that regulate developmental processes such as stem elongation and flowering. Fig. 5.1 shows the effect of different levels of GA on stem elongation. Plant 1 lacks GA and has an internode length of "0" ("dwarf" plant). Plant 2 has a moderate amount of GA and an average internode length ("normal" plant). Plant 3 has a large amount of GA and so has a much longer internode length ("giant" plant).

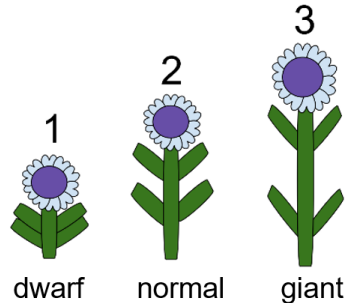


Fig. 5.1

Gene **K** encodes an enzyme responsible for one of the first steps of GA biosynthesis in *Arabidopsis* plants. The recessive allele of this gene results in GA-deficient *Arabidopsis* dwarfs. The dominant allele of Gene **T** codes for an activator which triggers the over-expression of Gene **K**, resulting in *Arabidopsis* giants.

When pure-breeding normal plants are crossed with pure-breeding dwarf plants, they produced F_1 which are all giants. Selfing of the F_1 generation produced the following F_2 generation:

443 giants
159 normal
197 dwarfs

- (a) Draw a genetic diagram to show the results of the F_2 generation from the selfing of the F_1 generation.

[3]

(b) (i) The expected phenotypic ratio of the F₂ generation is 9 giants : 3 normal : 4 dwarfs.

Calculate the value of χ^2 applicable to these data.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

where Σ = 'sum of...'

ν = degrees of freedom

c = number of classes

O = observed 'value'

E = expected 'value'

[2]

(ii) Use the calculated value of χ^2 and the table of probabilities provided to find the **probability** of the observed results differing by chance from the expected numbers.

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Probability of observed results differing by chance from expected numbers:

[2]

(iii) State the conclusions that may be drawn from the probability found in (ii).

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[Total: 10]

QUESTION 6

Bacteria can undergo genetic recombination, a process by which genetic information from one bacterium is transferred to, and then recombined with that of another bacterium.

Experiments have shown that two strains of bacteria can undergo conjugation when grown in a common medium. However, when they are separated by a filter in the Davis U-tube, as shown in Fig. 6.1, conjugation cannot occur.

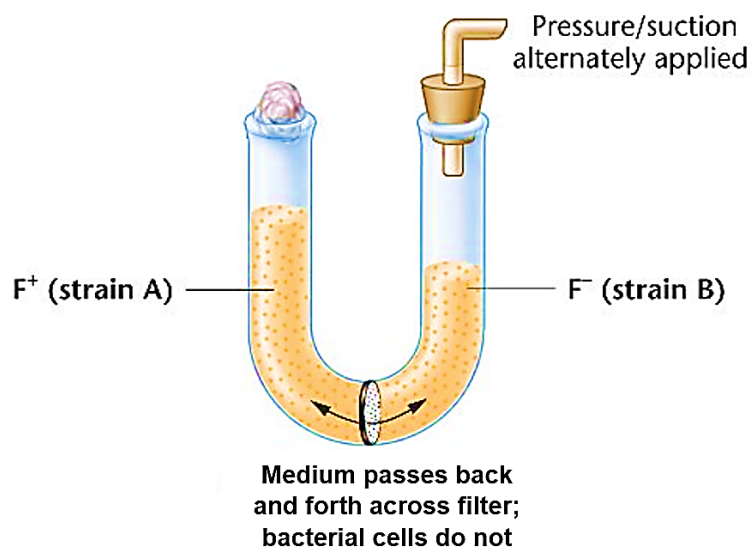


Fig. 6.1

(a) With reference to Fig. 6.1, explain why conjugation cannot occur when the two strains of bacteria are placed in the Davis U-tube.

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In another experiment investigating possible recombination in the bacterium *Salmonella typhimurium*, when researchers mixed *Salmonella* strains X and Y in a common medium, they recovered recombinants.

In a separate experiment using the Davis U-tube, the strains X and Y were separated by a filter, thus preventing cell contact but allowing growth to occur in a common medium. Surprisingly, when samples were removed from both sides of the filter, recombinants were recovered only from the side of the tube containing strain X bacteria. Researchers postulated that a filterable agent was released by the strain Y cells and responsible for transferring the new genetic information.

Three subsequent observations were useful in identifying the filterable agent:

1. The filterable agent was released by the strain Y cells only when they were grown in association with strain X cells.
2. The addition of DNase, which enzymatically digests naked DNA, did not render the filterable agent ineffective.
3. The filterable agent could not pass across the filter of the Davis U-tube when the pore size was reduced below the size of 20nm.

(b) (i) Suggest the source of the new genetic information which strain X cells received.

.....[1]

(ii) State the process of genetic recombination that has occurred.

..... [1]

(iii) Describe how the filterable agent could have transferred the new genetic information to strain X.

[3]

Fig. 6.2 represents the reproductive cycle that a T4 phage can carry out.

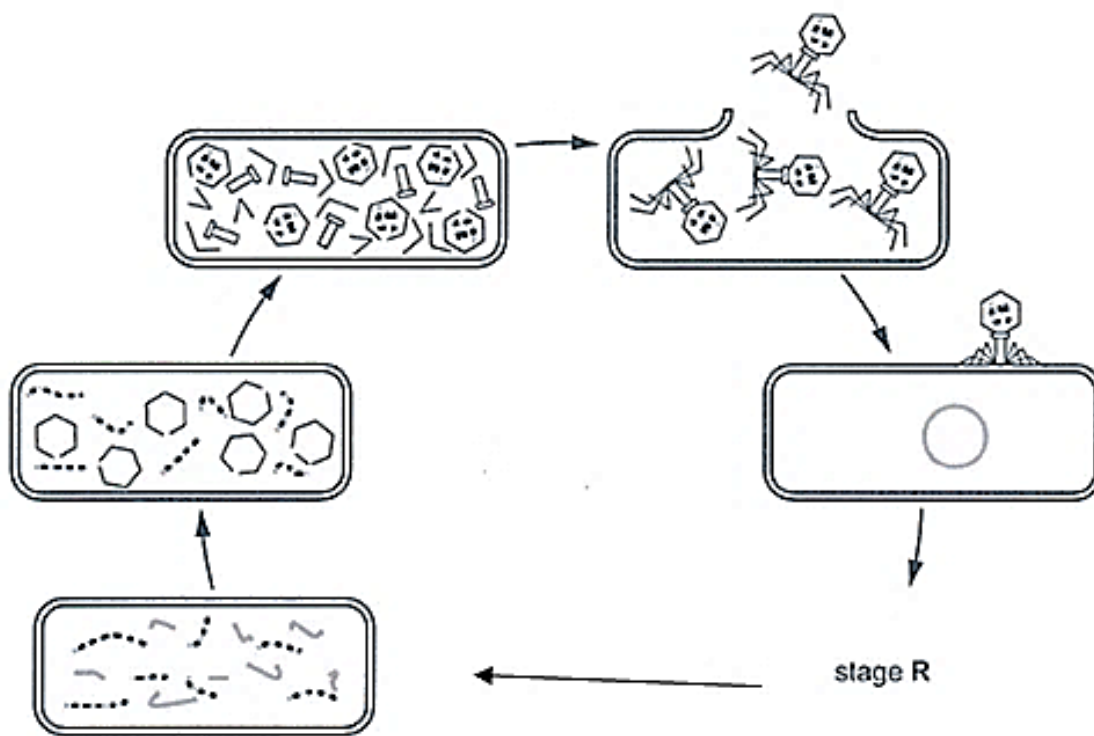


Fig. 6.2

(b) The algae that live within the cells of coral polyps have five different chloroplast pigments.

Table 7.1 shows the light wavelengths at which each algal chloroplast pigment shows its two largest peaks of light absorption.

Table 7.1

chloroplast pigment	peak 1 wavelength / nm	peak 2 wavelength / nm
chlorophyll a	430	662
peridinin	456	485
chlorophyll c ₂	450	396
dinoxanthin	442	471
β-carotene	454	480

Corals can be kept in glass tanks that are usually lit by lamps radiating mainly violet and blue light, with wavelengths in the range of 400 nm to 490 nm.

With reference to Table 7.1, suggest why lamps radiating mainly violet and blue light are expected to increase the growth of coral polyps more than lamps radiating light of all wavelengths.

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- (c) ATP is used or produced at different stages in the oxidation of glucose in aerobic conditions. Complete Table 7.2 to show whether ATP is used or produced at each stage of respiration. Write either **YES** or **NO** in each box.

Table 7.2

stage of respiration	ATP used	ATP produced
glycolysis		
link reaction		
Krebs cycle		
oxidative phosphorylation		

[2]

- (d) An experiment was carried out to investigate the effect of epicatechin on mitochondrial respiration in mice. Epicatechin is a naturally occurring compound in cocoa beans and so is present in chocolate. Two groups of mice, group A and group B, were used in this experiment.

- Group A was given water containing epicatechin, twice a day for 15 days.
- Group B was given water without epicatechin, twice a day for 15 days.

After 15 days, the structure of mitochondria from striated muscle cells in both groups of mice was examined. The surface area of the inner membrane of the mitochondria was divided by the surface area of the outer membrane to obtain a ratio for each mouse. Table 7.3 shows the mean ratios for the two groups of mice.

Table 7.3

group	mean ratio
A	2.0 : 1
B	1.7 : 1

The mice in group A were observed to be able to exercise longer than the mice in group B. With reference to Table 7.3, explain why the mice in group A were able to exercise for longer than the mice in group B.

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[Total: 10]

QUESTION 8

- (a) A geneticist studied the following pedigree diagram in Fig. 8.1 of a family with several members diagnosed with a genetic condition which originated from a mutation in a particular gene.

Given that the disease is autosomal dominant, Southern Blotting was also performed using genomic DNA isolated from each family member. The DNA samples were first subjected to restriction digestion by *EcoRI*, before hybridising to probes complementary to the said gene.

The results for all family members, except those for the foetus, are shown below in Fig. 8.1. The sizes of the DNA fragments obtained after *EcoRI* digest are as indicated.

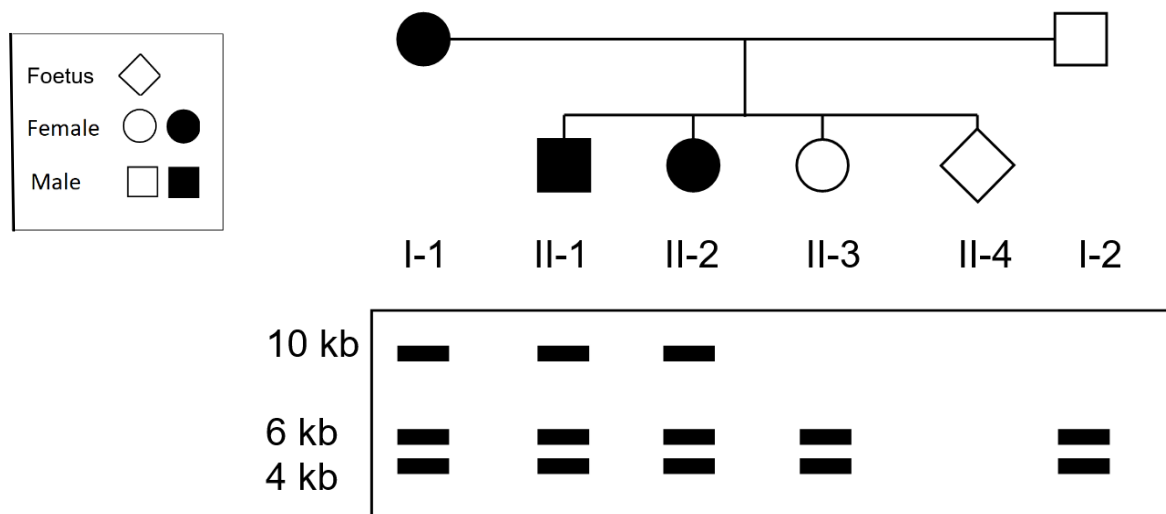


Fig. 8.1

- (i) With reference to the Southern Blot results of individual **I-1** in Fig. 8.1, use arrows to indicate the relative positions of the *EcoRI* restriction sites on the region of the genomic DNA recognised by the probe.

Label ALL the restriction sites as “**R**” and indicate the length of bases in kb.

allele 1 _____

allele 2 _____

[2]

- (ii) Calculate the probability of the foetus being a boy and inheriting the disease. Explain your answer.

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..... [2]

- (b) Fig. 8.2 shows the possible restriction sites of a gene and the results of four individuals after subjecting their genomic DNA to Polymerase Chain Reaction (PCR) and restriction enzyme digest.

The number below each band indicates the molecular sizes of the individual fragments.

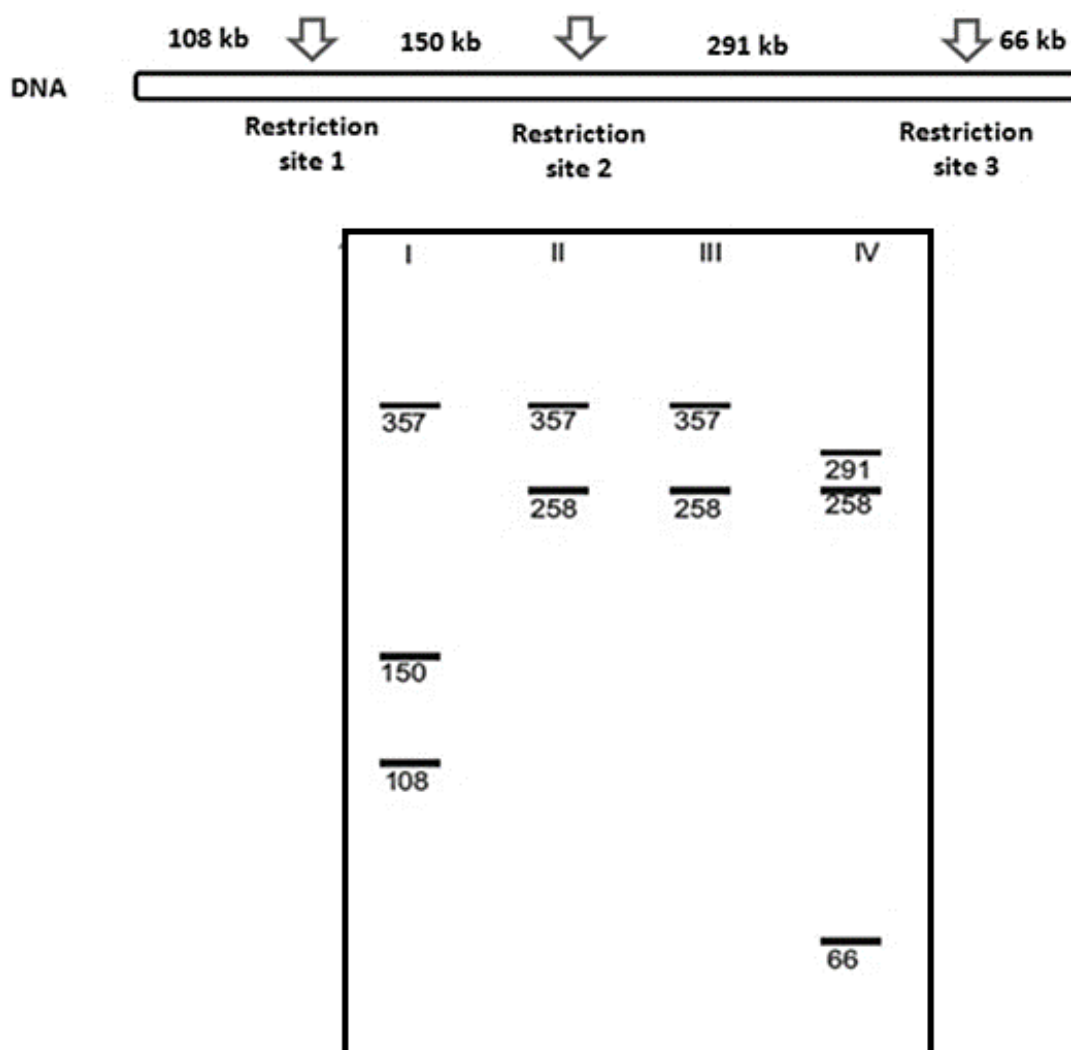


Fig. 8.2

- (i) Suggest the significance of carrying out PCR for the genomic DNA samples before subjecting the samples to restriction enzyme digest.

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- (ii) With reference to Fig. 8.2, identify the total number of alleles presented by this gene in the four individuals tested. Explain your answer.

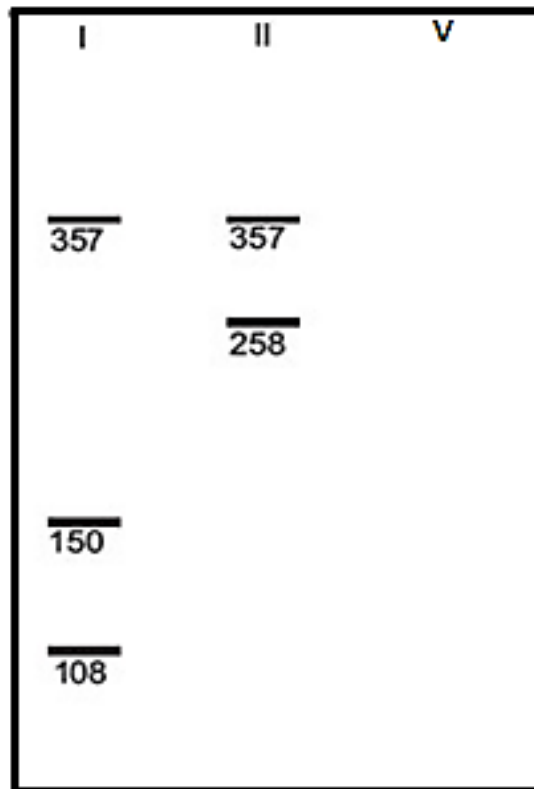
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- (iii) Suggest how a mutation can result in the different alleles for this gene.

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- (iv) Individuals I and II had a child V. Draw his banding pattern in the diagram below after subjecting his genomic DNA to the same PCR and restriction digest treatment.

[1]



[Total: 10]

QUESTION 9

The genus *Heliconius* contains more than 40 species of brightly patterned butterflies. Researchers have investigated in the laboratory how one species, *Heliconius heurippa*, could have developed as a separate species. The phenotype of *H. heurippa* is intermediate between that of two other species, *H. cydno* and *H. melpomene*.

Laboratory breeding experiments showed that:

- matings between *H. cydno* and *H. melpomene* (parent species) produce fertile hybrid offspring
- controlled matings of the hybrids produce individuals identical in appearance to *H. heurippa* within three generations
- hybrid butterflies prefer to mate with each other, rather than with individuals of either of the parent species.

(a) The researchers concluded that the *H. heurippa* species could contain DNA from the two parent species as a result of hybridisation.

- (i) Suggest, with reasons, one prediction that can be made about the chromosome numbers of *H. cydno* and *H. melpomene*.

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- (ii) The researchers thought that, because the hybrid butterflies preferred to mate with each other, this could make speciation more likely to occur.

Give reasons why the researchers thought that this made speciation more likely.

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- (b) *Heliconius* butterflies taste unpleasant to predators such as birds. The bright colours on the wings act as warnings so that birds avoid eating them.

Individual birds learn which patterns to avoid. If one *Heliconius* species is abundant, or if it has a pattern shared with another similar species, predators learn to avoid this pattern faster. Therefore, this pattern provides a selective advantage.

Fig. 9.1 shows the distribution of several *Heliconius* species in South America.

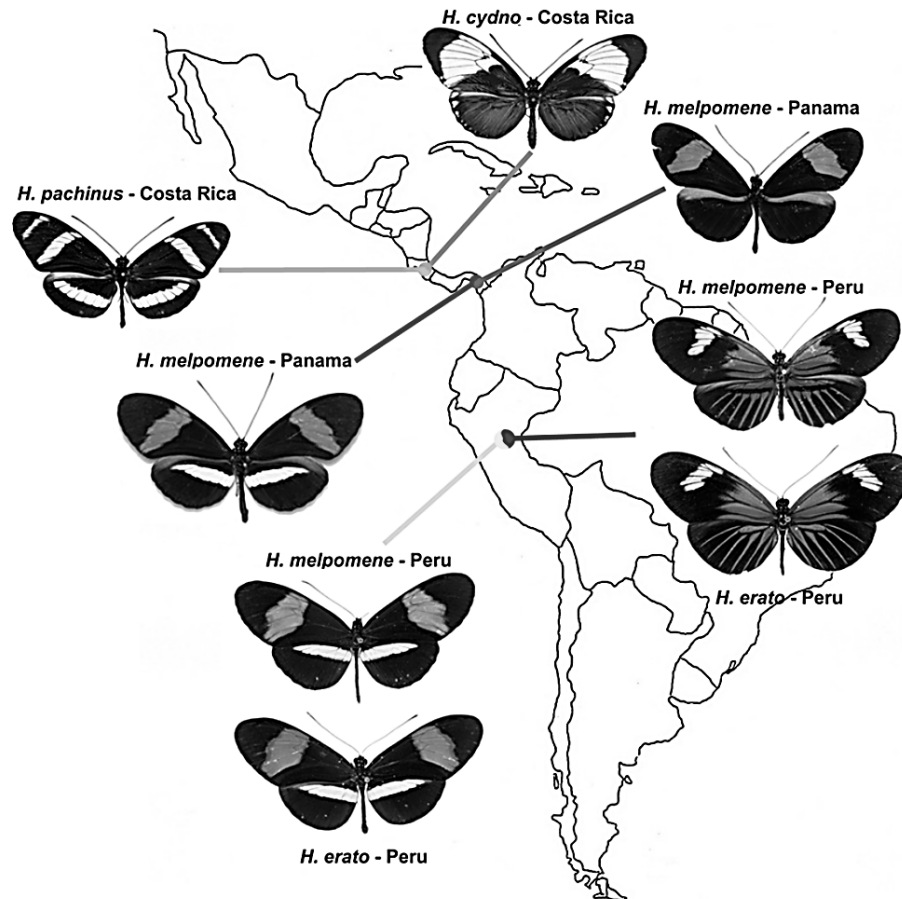


Fig. 9.1

With reference to Fig. 9.1 and your knowledge of anatomical homology and biogeography, comment on the evolutionary relationships between *H. Melpomene*, *H. cydno*, *H. pachinus* and *H. erato*.

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[Total: 8]

QUESTION 10

Fig. 10.1 shows some of the general changes in the immune system during a viral infection.

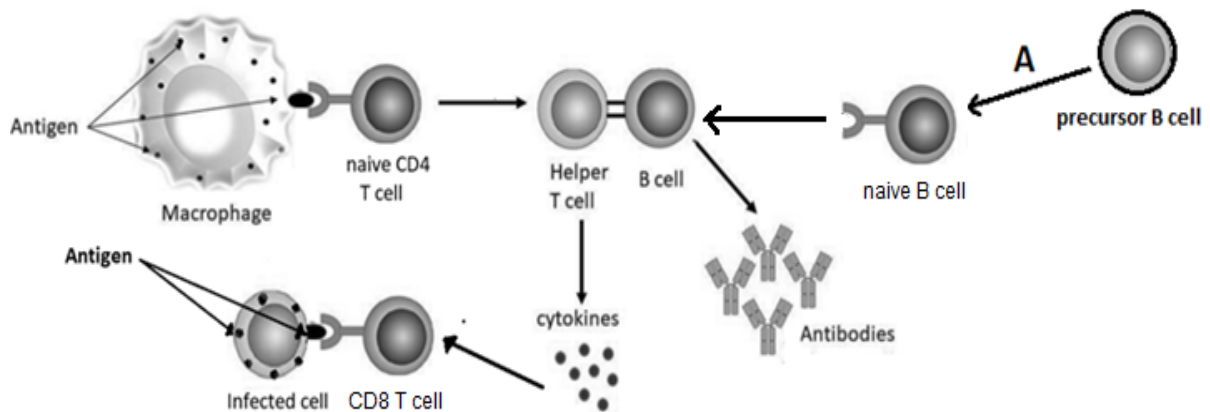


Fig. 10.1

- (a) Outline the genetic mechanism that occurs in stage **A** and explain its significance in the adaptive immune response.

[4]

- (b) A HIV-infected person infected saw his CD4 T cell count decrease within the first few weeks of his infection and then start to increase again.

With reference to Fig 10.1 and your own knowledge on adaptive immunity, suggest an explanation for the changes in his CD4 T cell count.

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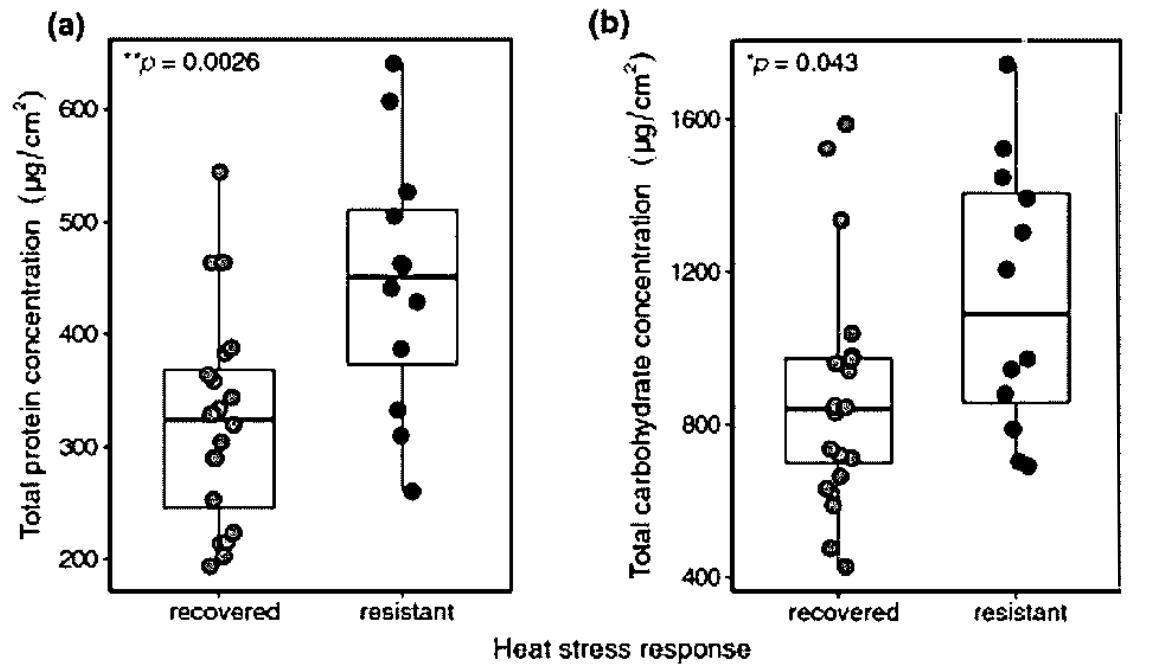
[Total: 6]

QUESTION 11

Rising seawater temperatures are contributing to coral bleaching, with mass coral bleaching events projected to increase in both frequency and severity.

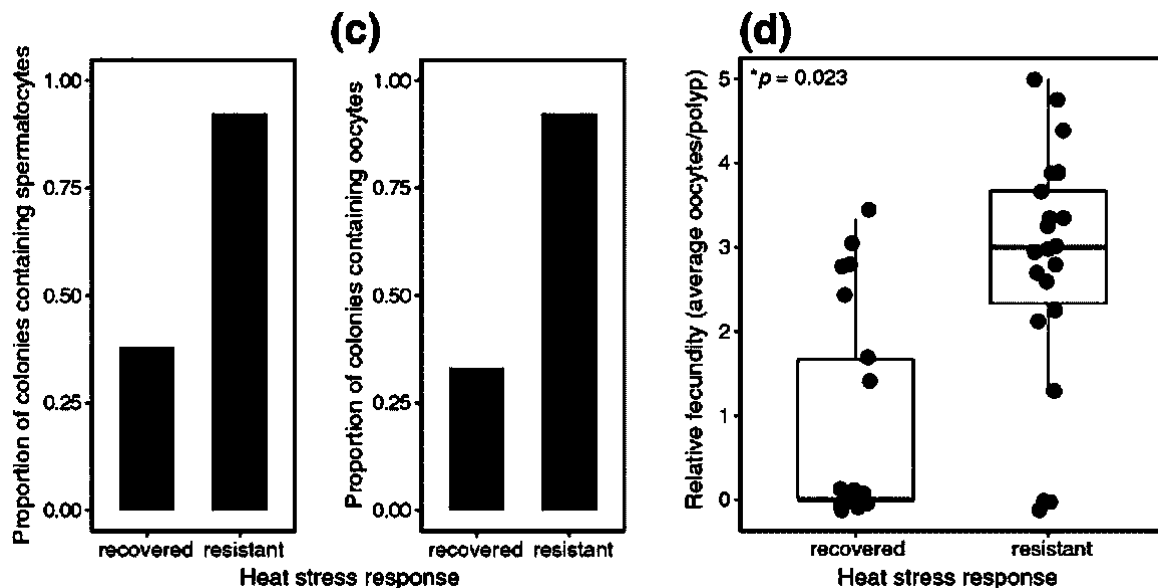
Fig 11.1 shows the impact of thermal bleaching stress on stored protein and carbohydrate reserves and reproductive output of two categories of corals five months after the bleaching event – corals that were resistant to bleaching events and corals that were bleached but later recovered.

For both types of corals, zooxanthellae was able to recolonise them and both types of corals also looked visually healthy.



Energetic condition of recovered and resistant coral colonies five months after mass bleaching event.

(a) Total protein content normalised to host tissue surface area. (b) Total carbohydrate content normalised to host tissue surface area. The data point represents a single colony.



(c) Proportion of recovered and resistant colonies containing spermatocytes and oocytes. (d) Relative fecundity (fertility) in recovered and resistant colonies. Each data point represents one colony.

Fig. 11.1

- (a) Suggest which type of corals would demonstrate better promote reef recovery after disturbance. Justify your choice with evidence from Fig. 11.1.

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Synthetic biology offers the potential to isolate and cultivate the resistant strains of coral that can naturally withstand higher sea surface temperatures associated with climate change.

Fig. 11.2 shows the process of using synthetic biology in promoting community reef recovery.

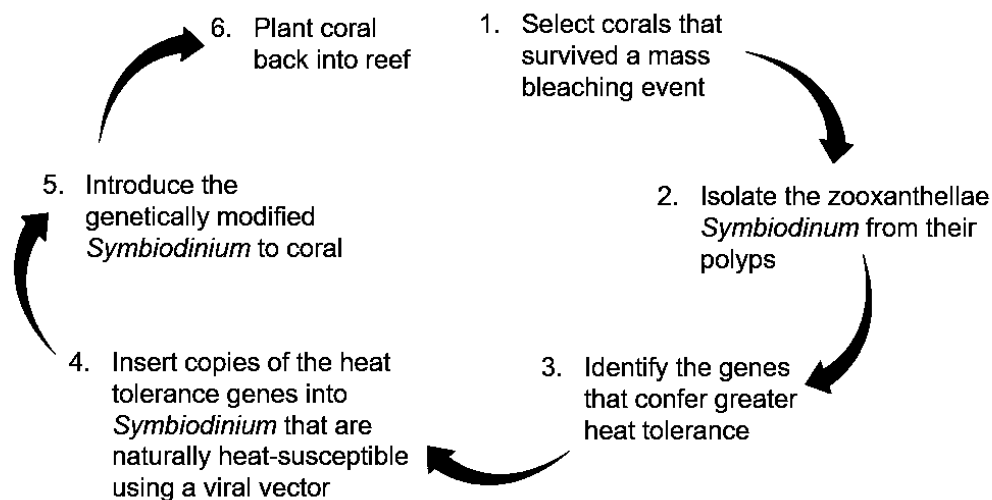


Fig. 11.2

A national survey was conducted with 1,148 Australians, measuring psychological predictors of support for a synthetic biology conservation solution to coral loss.

Participants were moderately strongly supportive of technology development and were most keen to implement genetically engineered coral with between 50 and 70% of reef remaining intact.

The findings suggest that the general public are not averse to the development of a synthetic biology solution for restoring the reef, and they may be especially influenced by whether the synthetic biology solution is shown to be effective, particularly in comparison to other conservation solutions (such as rubble stabilization, manual removal of coral predators like crown-of-thorns starfish).

However, support for a synthetic biology intervention is conditional and many participants expressed concerns about possible long-term impacts on humans, animals, and the environment as a result of deploying engineered coral.

- (b) Discuss the ethical considerations that support and oppose the use of genetically engineered coral for reef restoration.

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[Total: 5]

End of Paper